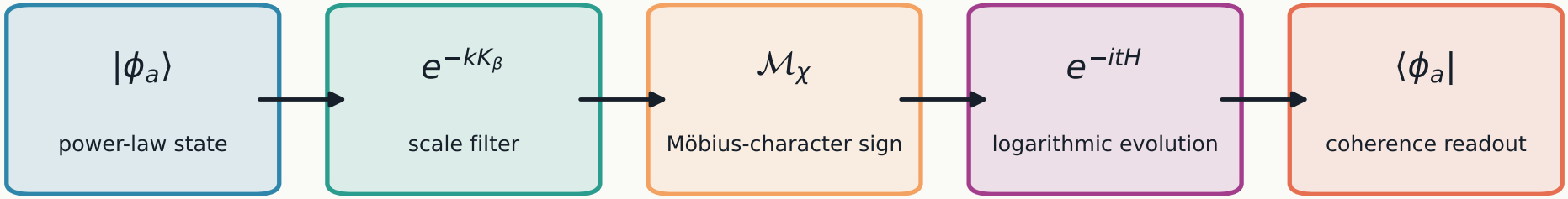
$$\mathcal{C}_{a,\beta,\chi}(k,t) = \langle \phi_a | \mathcal{M}_\chi e^{-kK_\beta} e^{-itH} | \phi_a \rangle$$



$H|n\rangle = \log n|n\rangle$ and $K_\beta = -\log(I - e^{-\beta H})$